

LESSONS LEARNED: THE INFLUENCE OF TESTING PROPERTIES ON UNIAXIAL COMPRESSION TESTS OF ICE

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ABSTRACT

This paper describes the influence of testing properties (test rig stiffness and PID controller settings) on the uniaxial compression test of granular laboratory-made ice in a single-factor design experiment. Global warming causes the decline of sea ice in the Arctic regions. Enabling the Northern Sea Route and the Northwest Passage to be more accessible for the shipping industry and therefore reduces the energy consumption compared to the Suez Canal Route. Nevertheless, ice will remain an interacting element with ships and structures in Polar and Sub-Arctic regions. A good assessment of forces exerted by the ice on the off-shore structure or the ship is essential for designing these arctic structures. Ice-structure interactions depend on the mechanical properties of the ice, which are still not fully explored but can be determined by experimental testing. Over the years, many experiments have been conducted with similar test setups but different testing properties. Therefore the influence of the test properties on the compressive strength of ice is analyzed and discussed by comparing the results of different test settings in this paper. The tests are conducted in a single-factor design experiment. The applied velocities range from 0.01 mm/s to 500 mm/s, causing the ice to behave ductile at lower velocities and brittle at higher velocities.

1 INTRODUCTION

In the Polar shipping areas of the world, the activity is increasing. The presence of sea ice is an additional constraint in the complexity of operations in Polar regions, and the compressive strength is a fundamental property of sea ice, among other properties. Observations of both large and small-scale sea ice failures show that ice often fails in compression [1]. Although the compressive strength has been extensively studied by Moslet in 2007 [2] among others, the uniaxial compressive strength is not easy to measure since it is challenging to induce a state of true uniaxial stress. In a conventional uniaxial compressive test of a solid material, an axial force is applied at a constant strain or stress rate to the ends of a right circular, cylindrical specimen until it fails. When compressed between steel platens, friction between the platens and the specimen produces radial restraint so that there is a triaxial state of stress near the end planes of the specimen; the triaxial field is significant over an axial distance of about one specimen radius from the end planes [3].

Ice shows in general two kinds of behavior under compression. It behaves ductile at low strain rates and brittle at higher strain rates. The growth and interaction of cracks control brittle behavior [4]. When ice behaves brittle, it will fail at the end of a uniaxial test, whereby the failure mode is influenced by the boundary conditions at the end of the specimen [5]. When using synthane end-caps, the boundary conditions increase the brittle uniaxial compressive strength of the ice specimen, mainly failing by shear faulting. At higher confinement, the ice fails by spalling [4]. Without this confinement, the ice specimen is likely

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to fail by axial splitting resulting in lower compressive strength values [4, 6]. In addition, there are other influencing factors either intrinsic (temperature, salinity, density, ice type, crystal size and orientation) or related to test properties (loading rate, loading direction, sample size, stiffness of the test rig, and sample preparation techniques) [1].

Haynes and Mellor in 1977 showed that cylindrical specimens of ice may be tested down to a L/D ratio of 0.74 with no apparent effect upon the compressive strength of ice when suitably designed compliant platens are employed [3]. Otherwise, the compressive strength increases with decreasing L/D ratio, which was also reported by Hawkes and Mellor in 1970 [7]. Strain rate calculations were based on machine speed and time to failure [3]. Unfortunately, Haynes and Mellor did not specify the failure mode. Schulson described in 1987 [6] the fracture of ice under uniaxial compression in terms of fracture toughness, end constraint, coulombic friction, and crack interactions within a shear fault. The decrease in compressive strength between 10^{-3} 1/s and 10^0 1/s (by about 30%) was explained by Schulson in terms of the strain- or sliding-rate-dependent decrease in coulombic friction at already formed crack surfaces sliding against each other [6]. Jones in 1997 conducted high-speed uniaxial compression tests of columnar grained ice in the strain rate range 10^{-1} to 10^1 1/s and observed an increase in the uniaxial strength of ice specimens with a L/D ratio of 2.08 at increasing strain rates [8] without indicating the failure mode. Shazly et al. in 2009 investigated the dynamic response of single- and polycrystalline (columnar) ice under uniaxial compression for strain rates of 60 to 1400 1/s and showed a positive strain-rate sensitivity of the compressive strength of ice specimens with a L/D ratio of 2 to 3 [9]. Lian et al. [10] investigated cubic columnar ice specimens ($L/D = 1$) at an apparent engineering strain rate range of 2 to 400 1/s and showed that there exists a significant difference between actual engineering strain rate and apparent engineering strain rate. In addition, the data showed that the uniaxial compressive strength of lake ice has a positive strain rate sensitivity when splitting failure occurs. Qi et al. [11] thereby presented a series of uniaxial compressive experiments on natural (columnar) lake ice specimens ($L/D = 2.5$) at strain rate range of 10^{-1} to 10^2 1/s near melting temperature (-0.1°C) and also show a positive strain rate sensitivity of the compressive strength at a splitting failure mode.

The authors utilized granular laboratory-made freshwater ice specimens. A 160 kN hydraulic testing machine was employed for the loading process with external displacement control. The apparent engineering strain rate ranges from 4×10^{-5} to 4×10^0 1/s at testing temperatures of -10°C . The influence of the test rig stiffness and the PID controller (proportional-integral-derivative controller) settings on the actual engineering strain rate at the uniaxial compression test of ice over the elevated strain rate were investigated. At last, the data in the present study were compared to previous uniaxial compression tests in the literature.

2 EXPERIMENTAL PROCEDURES

The conducted tests comprised four series of uniaxial ice compression tests of granular ice. All tests were performed at the Hamburg University of Technology in the mechanical laboratory of the Institute for Ship Structural Design and Analysis.

2.1 Specimen production

The ice specimens were cylindrical with a diameter of 99.4 mm. The total length of the specimen was 250 mm resulting in a L/D ratio of 2.5, which is an ideal ratio as Schwarz et al. suggested [12]. The ice specimens were frozen using two thirds commercially available crushed ice and one third distilled water (not deaerated) with a mean acidity of $8.6 \text{ pH} \pm 1 \text{ pH}$ and a mean electrical conductivity of $7.5 \mu\text{S/cm} \pm 5.8 \mu\text{S/cm}$ measured at temperatures of $18.8^\circ\text{C} \pm 3.6^\circ\text{C}$. The added water was cooled down to 5°C before use to prevent excessive melting of the crushed ice. The ice specimens were frozen in PVC-U pipes at -10°C in a refrigerated container. To avoid cracks in the ice specimens, they were frozen from the bottom up by adding a thin aluminum plate to the bottom of the PVC-U pipe, directly standing on the floor in the refrigerated container. The top of the PVC-U pipes was covered with insulating material. This procedure led to a relatively homogeneous granular grain structure over the length of the sample with more significant differences at both ends of the specimen. A cross-section was taken from the center of a random specimen, as shown in Figure 1.

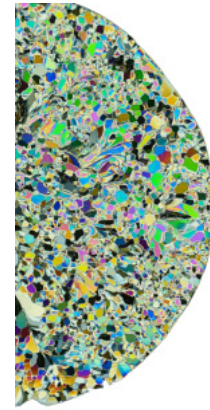


FIGURE 1: Typical microstructure of a specimen

After freezing for at least two days, the PVC-U pipe was removed by heating it up with a hot air gun (about 2000 W) for a short time. Immediately after removal, the ice specimens were stored in the refrigerated container. Each specimen was cut, at both ends, into the length of 250 mm with a -10°C cooled band saw. This process ensured consistency in the shape and material properties of the specimens. In total, 123 specimens in four series

were produced and tested. The specimen production order was randomized to distribute uncontrollable factors (such as temperature changes, e.g., in the environment) within the experimental plan in each manufacturing series.

2.2 Setup and Methods

The equipment used in this test series includes a hydraulic cylinder, a load cell, an external displacement sensor, a high-speed camera and a second refrigerated container, see Figure 2.

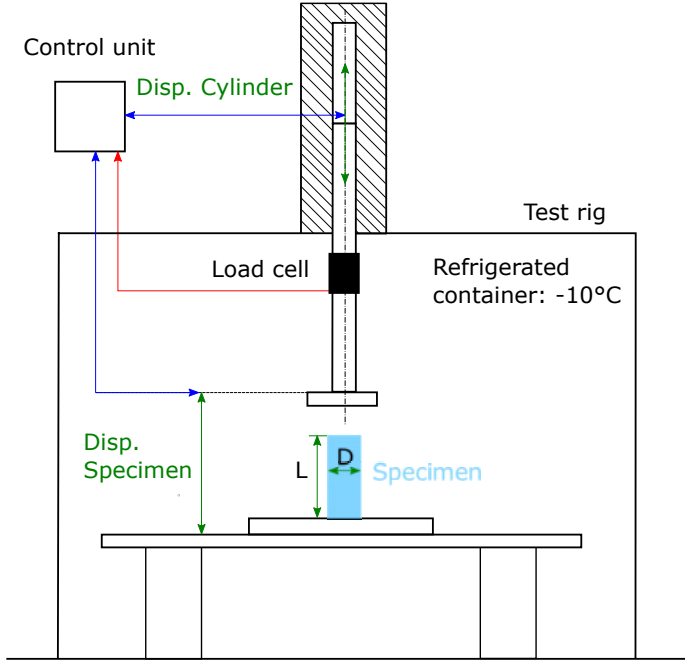


FIGURE 2: Test setup

The green arrows show the displacement measurements, the red arrow shows the direction of the load signal, and the blue arrows show the direction of the displacement signals. The hydraulic cylinder was a *Schenck Hydropuls-Longitudinal PL 160* cylinder with a 160 kN nominal load, which used a build-in inductive displacement sensor. The *Schenk PM160Rn* load cell was attached between the hydraulic cylinder and the stamp. The test was controlled with the *Instron Labtronic 8800* and the measurements were carried out with the *Gantner Q.raxx-station 101 T* measuring amplifier. The displacement, the force and the temperature were measured with the *Gantner Q.raxx A101* module. The measured surface roughness of the stamp and the bottom platen according to the ISO 1997 was $R_z = 8.7 \mu\text{m} \pm 2.25 \mu\text{m}$ for $\lambda_c = 0.8 \text{ mm}$. In addition, an external displacement sensor (*burster 8718-500*) was physically connected to the stamp in order to measure the displacement of the specimen. The build-in

inductive displacement sensor was located at the head end of the cylinder approximately 1 m above the specimen. For test velocities up to 100 mm/s the data acquisition rate was 10 kHz. In the forth series (velocities above 100 mm/s) the data acquisition rate was 100 kHz. High-speed recordings of the ice specimens under compression were made to record the failure behavior. The recording rate was 500 to 2500 frames per second. All specimens failed ductile or brittle depending on the strain rate. The transition region could not be hit, and therefore failure in the transition zone could not be achieved.

3 ANALYTICAL ASSESSMENT OF THE TRUE LOAD VELOCITY

During a displacement-controlled compression test, the cylinder force (or a comparable actuator) generates a deformation of the specimen and the surrounding test structure. For an ideally rigid test structure, the cylinder velocity can be assumed to be equal to the loading velocity. However, compression specimens are relatively stiff, so this assumption does not generally apply. Therefore, the influence is to be estimated analytically. The total stiffness of a compression test could be idealized by a summation of the stiffnesses of the specimen and the test rig.

$$\frac{1}{C_{Total}} = \frac{1}{C_{TestRig}} + \frac{1}{C_{Specimen}} \quad (1)$$

$$C_{Total} = \frac{F_{Cyl}}{\Delta s_{Cyl}} = \frac{F_{Cyl}}{v_{Cyl} \cdot t} \quad (2)$$

$$C_{Specimen} = \frac{E \cdot A}{L} = \frac{F_{Specimen}}{v_{Specimen} \cdot t} \quad (3)$$

According to equilibrium of forces $F_{Cyl} = F_{Specimen}$ and together with Equation 2 and 3 the following applies.

$$C_{Total} \cdot v_{Cyl} \cdot t = C_{Specimen} \cdot v_{Specimen} \cdot t \quad (4)$$

Together with Equation 1, the equilibrium of forces leads to the following formula for estimating the true loading velocity applied to the specimen $v_{Specimen}$.

$$v_{Specimen} = v_{Cyl} \frac{C_{TestRig}}{C_{TestRig} + C_{Specimen}} \quad (5)$$

The results of this relation are presented normalized in Figure 3. If the stiffness of the test rig is 10 times greater than that of the specimen, the deviation is about 9%. For a ratio of 100, the deviation is even less than 1%. For stiffness ratios of 1, as in the case of the present study (stiffness of the test rig was around 65 kN/mm according to a FEM simulation), the deviation is 50%.

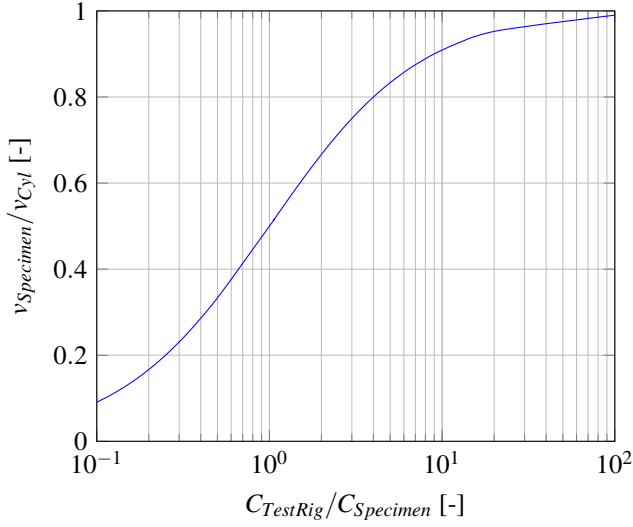


FIGURE 3: Ratio between cylinder- and true load velocity in dependence of the ratio of specimen- and test rig stiffness.

Since it is often difficult to increase the stiffness of the test rig or change the specimen geometry, the deviation of the loading velocity must be taken into account for the control and evaluation of the test.

4 EXPERIMENTAL RESULTS

The experimental plan consists of 123 specimens tested in four series. Table 1 shows the velocity levels in each series.

TABLE 1: Incomplete block design [13] for uniaxial compression tests

Series	Treatments (Levels)			
1	0.01 mm/s	0.05 mm/s	0.06 mm/s	1 mm/s
2	1 mm/s	10 mm/s	100 mm/s	
3	0.1 mm/s	1 mm/s	10 mm/s	100 mm/s
4	1 mm/s	10 mm/s	100 mm/s	500 mm/s

Commonly, the compressive strength of ice is given over the apparent engineering strain rate [3, 10]. The compressive strength of ice of the present study is given here over the actual engineering strain rate.

4.1 Definition of the actual engineering strain rate

The velocity during compression v divided by the specimen's original length L defines the strain rate and the maximum measured force $F_{\max}$ divided by the cross sectional area A of the specimen defines the compressive strength, see Equations 6 and 7.

$$\dot{\epsilon} = \frac{v}{L} \quad (6)$$

$$\sigma_c = \frac{F_{\max}}{A} \quad (7)$$

At the beginning of the test, the experimenter sets the cylinder's nominal velocity, which then defines the apparent engineering strain rate ($\dot{\epsilon}_n$). The actual velocity is needed for the actual engineering strain rate ($\dot{\epsilon}_t$), but it is not entirely trivial to determine the actual velocity in an experiment. The authors create a linear regression over the measured displacement signal, and the slope of the regression curve is then defined as the actual velocity. Before doing the regression, the data is cut off as follows: Starting point is a force of 1 kN and the maximum force reached during the test defines the endpoint. Figure 4 shows the measurement of the displacement of the specimen during the compression test in series 4 run number 26 (S4No26) at a nominal velocity of $v_n = 100$ mm/s.

The coefficients are determined with 95% confidence bounds ($R^2 = 0.993$), and the actual velocity of test number 26 of series 4 is $v_n = 83.75$ mm/s, which deviates from the apparent velocity by 16%. This deviation of the actual engineering strain rate to the apparent engineering strain rate is influenced by the test rig stiffness and the PID controller settings, among other influences, and will be discussed in the following.

4.2 The influence of the test rig stiffness

A stiffness ratio above 100 between the test rig to the specimen under investigation is preferable, see Section 3. Otherwise relative movement between the test rig and the specimen occurs during the compression test. Unfortunately, such a ratio is difficult to achieve and difficult to modify. The specimen displacement control (SDC) of the test, in which the cylinder is controlled via the displacement measurement of the specimen, provides a remedy for the influence of the test rig stiffness. Figure 5 shows the force and displacement curves of two ductile tests.

Both tests were performed at the same nominal velocity of 0.1 mm/s. The displacement is shown on the left ordinate, and on the right, the force is shown. The time during measurement is on the abscissa. The measurement of the displacement on the specimen is blue, and the measurement of the cylinder displacement

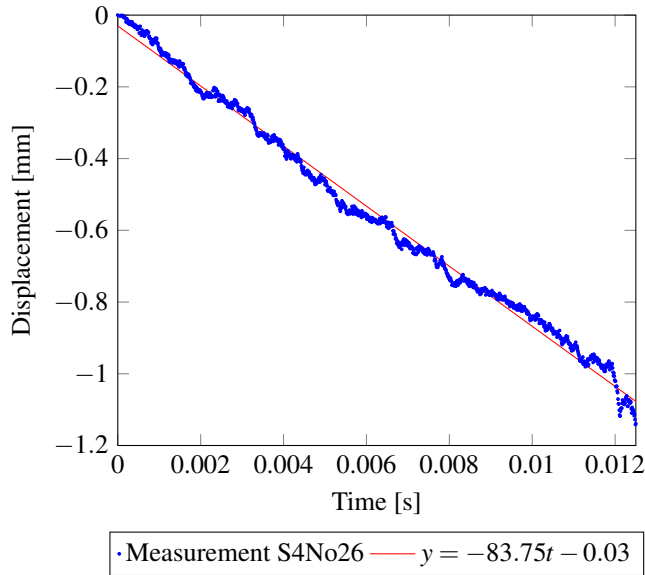


FIGURE 4: Displacement measurement and linear fitting; Run number 26 series 4

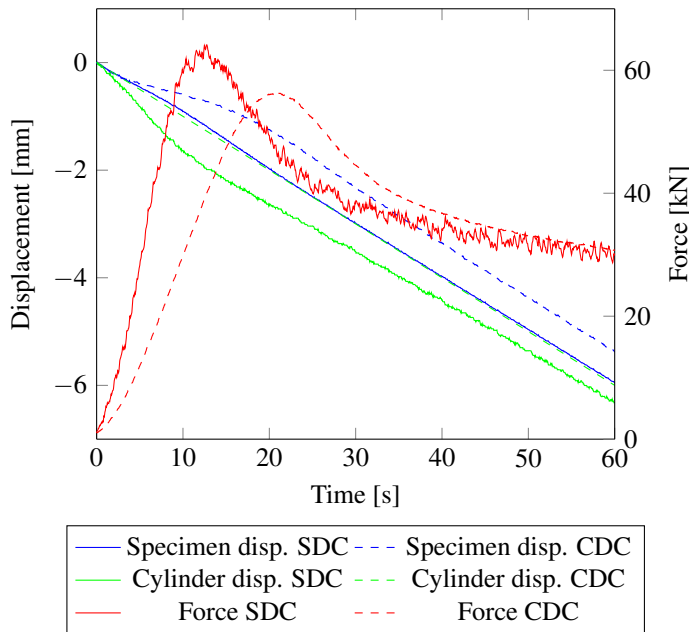


FIGURE 5: Displacement and force measurements (ductile behavior)

is green. In red, the force is shown. The test with the cylinder displacement control (CDC) is shown with dashed lines and the test with the specimen displacement control is shown with full lines. When looking at the course of the displacement measurement of the cylinder displacement controlled experiment, it can

be seen that the cylinder reaches approximately the desired velocity, but the velocity at the specimen decreases above a certain force. This can be attributed to the relative displacement/motion of the test rig. The slope of the displacement measurement of the specimen displacement controlled experiment shows compensation for this effect. The cylinder starts to accelerate to maintain the speed at the specimen. In series 3, the PID components of the control have been sufficiently adjusted. The deviation from the apparent strain rate is 1% for the specimen displacement controlled test and 3% for the cylinder displacement controlled test. However, this difference is not negligible for tests in the ductile range since the tests usually have a long test time, and therefore the actual engineering strain rate has a significant influence on the results. In the first series of tests, the PID control settings were set to default and specimen displacement controlled ductile tests at a rate of 0.01 mm/s resulted in a relative deviation of the actual velocity of 1%, whereas the deviation was up to 13% for cylinder displacement controlled tests. It has been shown that specimen displacement control (SDC) is preferable to cylinder displacement control (CDC) for ductile tests.

4.3 The influence of the PID controller settings

The PID controller consists of the components of the P-element, the I-element, and the D-element. The P-controller consists exclusively of a proportional part of the amplification and reacts immediately. However, its use is minimal because the amplification must be strongly reduced depending on the behavior of the controlled system. The I-controller is an accurate but slow controller due to its (theoretically) infinite amplification. The D-element is a differentiator which is only used as a controller in conjunction with controllers with P- or I-behavior. Figure 6 shows the deviation between the actual and apparent displacement plotted over the force of test S1No9 (circular marker) and test S2No13 (square marker) with adjusted PID controller settings.

Both tests were performed at the same apparent velocity of 1 mm/s with the specimen displacement control. The PID controller settings of test S1No9 were on default ($P = 15$ dB, $I = 1$ 1/s and $D = 0$ ms) and test S2No13 has the adjusted settings of $P = 22.7$ dB, $I = 0.1$ 1/s and $D = 1.5$ ms. In the test S2No13 with the adjusted settings it can be seen that the cylinder accelerates so that the velocity at the specimen approaches the apparent velocity. When testing with the default setting, the cylinder does not have enough time to adjust its velocity so that the velocity at the specimen strongly deviates from the apparent velocity. The relative deviation of the actual velocity ($v_t = 0.35$ mm/s) for the test with the default settings (S1No9) is 65% and the relative deviation of the actual velocity ($v_t = 0.94$ mm/s) for the test with the adjusted settings (S2No13) is 6%. The latter emphasizes using the actual engineering strain rate as it suggests a better correlation to the uniaxial compressive strength than the apparent strain

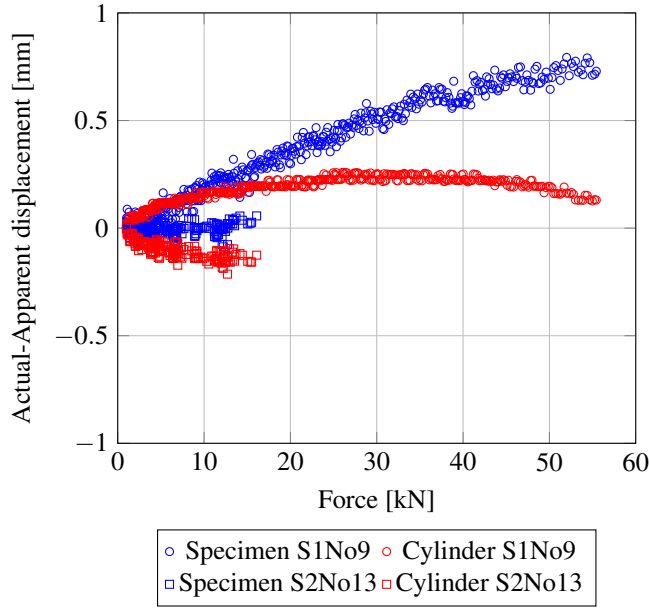


FIGURE 6: Deviation actual and apparent displacement vs. force for two different PID settings (brittle behavior)

rate. Furthermore, the adjustment of the PID components of the control has a significant influence on the actual velocity and becomes more important with increasing test velocities.

4.4 Results

Figure 7 shows the results of the uniaxial compressive test of granular ice in the first testing series.

The strain rate is depicted on a logarithmic scale, and the grid lines on the abscissa indicate the apparent strain rate. The red crosses show the specimen displacement controlled tests, and the blue circles show the cylinder displacement controlled test. For the strain rates tested up to 2.4×10^{-4} 1/s, the actual engineering strain rate is almost at the apparent engineering strain rate. However, it can be seen that the specimen displacement controlled tests are significantly closer to the apparent engineering strain rate. Up to this rate, the failure behavior is ductile. At apparent engineering strain rates of 4×10^{-3} 1/s, the difference between the apparent and actual engineering strain rate is more significant, whereas only one reference value was generated for cylinder displacement control. At this apparent engineering strain rate, brittle failure behavior is observed. Apart from the two outliers at an apparent engineering strain rate of 4×10^{-3} 1/s, a trend can be observed that the compressive strength of ice increases with increasing strain rate. The PID settings were on default.

The brittle failure process was captured with images from the high-speed camera. Figure 8 shows the failure process of the

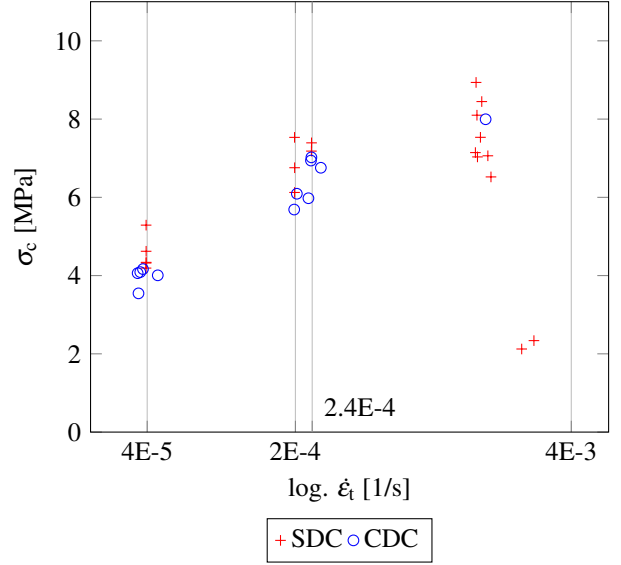


FIGURE 7: Uniaxial compressive strength of granular ice; series 1

first ice specimen tested at an apparent strain rate of 4×10^{-3} 1/s.

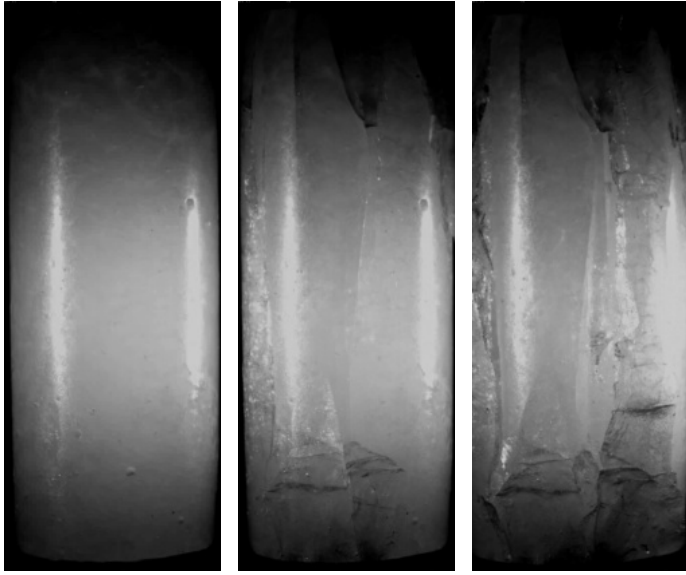
Figure 8a shows the intact specimen at the start of the compression test. Within a few milliseconds, first cracks develop on both ends of the specimen and extend toward the middle, see Figure 8b. The stress increases quickly, so does the number of cracks. Figure 8c shows that axial cracks may form as the cracks from the upper and lower end meet. As the compression continues the number of axial cracks increases and the specimen begins to separate into a few large ice columns, see Figure 8d, which will fracture through buckling (Figure 8e) and finally fails as the maximum stress is reached, see Figure 8f. All tested specimens that exhibited brittle behavior failed due to axial splitting.

Figure 9 shows the results of the uniaxial compressive test of granular ice in the second testing series.

Since the specimen displacement control led to better results in the first test series, only the specimen displacement control was used in the second series (red crosses). In addition, in the second series the PID setting had been adjusted to $P = 22.7$ dB, $I = 0.1$ 1/s and $D = 1.5$ ms with the trial and error method. Here it can be seen that the actual engineering strain rate is closer to the apparent engineering strain rate of 4×10^{-3} 1/s than in the previous series of tests. Even at ten times the strain rate, the ratio is still satisfactory. The difference becomes more pronounced at one hundred times the strain rate. For the tested strain rates of series two, it can be observed that the compressive strength of ice decreases with increasing strain rate.

Figure 10 shows the results of the uniaxial compressive test of granular ice in the third testing series.

The red crosses show the specimen displacement controlled



(a) Frame 0; Intact specimen (b) Frame 100; Development of first cracks at both ends (c) Frame 125; Development of axial cracks



(d) Frame 175; Development of ice columns (e) Frame 250; Fracture of ice columns (f) Frame 700; Specimen after the maximum stress is reached and their separation

FIGURE 8: Failure mode: Axial splitting at 1 mm/s

tests, and the blue circles show the cylinder displacement controlled test. In series 3, the test levels from series 1 and 2 were repeated in order to be able to make more statistically significant statements by keeping the adjusted PID settings of series 2. It can also be seen, that the specimen displacement control is more

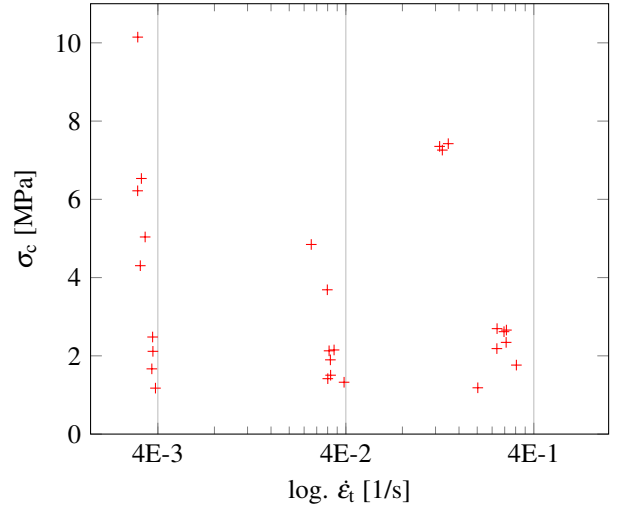


FIGURE 9: Uniaxial compressive strength of granular ice; series 2

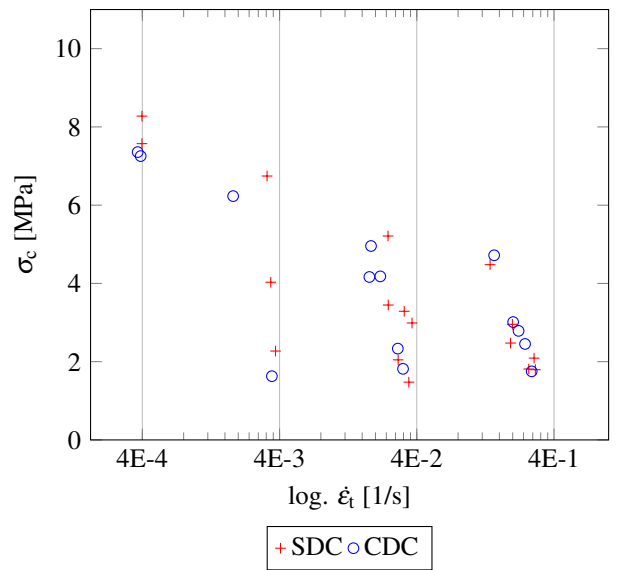


FIGURE 10: Uniaxial compressive strength of granular ice; series 3

advantageous than the cylinder displacement control at speeds of up to 10 mm/s ($\dot{\epsilon}_n = 4 \times 10^{-2}$ 1/s). Within the tests of series 3, the compressive strength of ice decreases with increasing strain rate.

Figure 11 shows the results of the uniaxial compressive test of granular ice in the forth testing series.

The aim of this test series was to test up to the maximum rate of the hydraulic system, which was 1,000 mm/s. At these rates, the specimen fails within a few microseconds, so the mea-

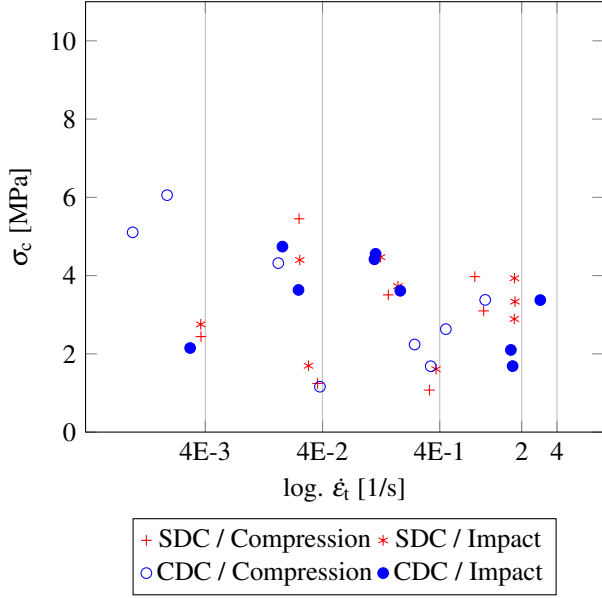


FIGURE 11: Uniaxial compressive strength of granular ice; series 4

surement frequency was increased to 100 kHz in this test series. After the first test at an apparent velocity of 1,000 mm/s, the system failed electronically, which was the reason that the maximum apparent rate was at 500 mm/s in the further course. In addition, the apparent velocity of 1,000 mm/s could not be reached, and the actual velocity was 720 mm/s ($R^2 = 0.954$). The 1,000 mm/s test was conducted with a starting gap of 10 mm between the stamp and the ice specimen to ensure the correct speed before contact between platen and specimen. Because of the gap, the experimenter decided to label these kinds of tests 'impact', see Figure 11. It can be seen, that at an apparent strain rate of 4×10^{-3} 1/s, only the cylinder displacement controlled tests do not reach the apparent strain rate. However, at an apparent strain rate of 2 1/s, it can be seen that only the impact tests still approximately reach the apparent velocity. For the tested strain rates of series four, it can be observed that the compressive strength of ice decreases with increasing strain rate.

5 DISCUSSION, CONCLUSIONS AND RECOMMENDATIONS

In the course of 123 uniaxial compression tests of granular ice specimens with an L/D ratio of 2.5 and at a test temperature of approximately -10°C , the influence of test properties was investigated. The following can be concluded:

1. A dependency between the control type (CDC vs. SDC) and the actual engineering strain rate could be shown up to a strain rate of 2×10^{-2} 1/s. The cylinder displacement con-

trol deviates more from the apparent velocity than the specimen displacement control, strongly depending on the stiffness of the testing rig, see also Section 3. The specimen displacement control brings a remedy by eliminating the influence of the stiffness of the test rig. Before testing, the experimenter should consider a gap between the stamp and the specimen at higher strain rates.

2. The PID controller setting also significantly affects the actual engineering strain rate and should be satisfactorily adjusted with preliminary tests.

Additionally, the data published here show a decrease in compressive strength between strain rates 10^{-3} and 10^0 1/s. In Figure 12 the uniaxial compressive strength of granular ice of the present investigation, besides some other studies, is shown over the actual engineering strain rate.

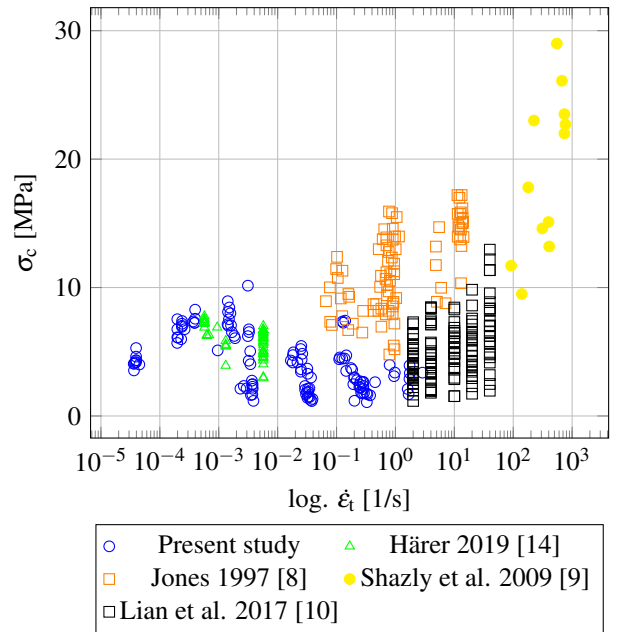


FIGURE 12: Uniaxial compressive strength of granular ice over actual engineering strain rate; Present study and others

The strength values in the brittle range are far below the strength values of Jones 1997 [8] and Shazly et al. 2009 [9]. The difference to Jones may be caused due to the different L/D ratio and the different grain type, whereby Shazly has investigated single- and polycrystalline ice at much higher strain rates. The data of [14] and [10] were given over the apparent strain rate, which has a distortion effect as this study shows, by which the actual engineering strain rates could deviate about 50% depending on the testing properties. In the present study, ice specimens showing brittle behavior failed through axial splitting. However,

Schulson [6] stated that the uniaxial strength seems to be limited by the shear faulting.

Comparing different experimental ice compression studies is challenging as testing properties, ice specimen geometries, and grain types differ, and the indication of the fracture mode is lacking. A first step in *establishing a common database of ice experiments and using machine learning to understand and predict ice behavior* was performed by Kellner et al. [15, 16] whereby the authors of the present study suggest that the brittle fracture mode should be categorized.

The data presented will be published in full later in the journal *Data in Brief* [17]. The full data can be accessed by the *Mendeley Data* repository [18], and thus the ice experiment database has been carefully expanded since data models can only be as good as the data on which they are based.

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